

Loss Investigation of Interior Permanent-Magnet Motors Considering Carrier Harmonics and Magnet Eddy Currents

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Abstract—In this paper, we investigate losses of interior permanent-magnet motors driven by pulsewidth-modulated inverters using 3-D finite-element analysis, which can estimate the eddy-current loss in the permanent magnet accurately. The calculated losses are compared with the measured results and the theoretical solution to verify the validity of the analysis. The variation of the magnet eddy-current loss due to the division of the magnet is also investigated. It is clarified that the eddy-current loss in the permanent magnet is mainly produced by the carrier harmonics of the inverter and that the axial length of the divided magnet should be smaller than the skin depth of the eddy currents produced by the major harmonics for the loss reduction.

Index Terms—Carrier harmonics, finite-element analysis, interior permanent-magnet synchronous motor, iron loss, pulsewidth-modulated (PWM) inverter.

I. INTRODUCTION

INTERIOR permanent-magnet motors driven by pulsewidth-modulated (PWM) inverters are widely applied to the industry [1]–[3]. The motors have many advantages. The speed range can be extended using the flux-weakening control. The reluctance torque can be utilized as well as the magnet torque. As a result, the motors can be driven at wide speed range with high efficiency.

On the other hand, the electromagnetic field in the motor includes many harmonics, for instance, the slot harmonics and carrier harmonics. They cause considerable harmonic losses at the stator and the rotor. In particular, in the case of the rare-earth magnets, the carrier harmonics produced by the PWM inverter may cause relatively large harmonic eddy-current losses because the conductivity of the magnet is large, and frequency of the carrier is over 1 kHz.

Reference [4] represented the iron losses of the interior permanent-magnet motors, considering the carrier harmonics of the inverters using 2-D nonlinear time-stepping finite-element

analysis. However, the 3-D analysis is required to take the correct path of the eddy currents in the magnet into account. It must be important to understand the proper division of the magnet pieces to prevent the heat demagnetization.

From these points of view, we investigate the losses of the interior permanent-magnet motors considering the magnet eddy currents and carrier harmonics using the 3-D nonlinear time-stepping finite-element analysis. First, the calculated results are compared with measurement to verify the validity of the analysis. The losses are decomposed into several components due to their origins, for instance, stator-slot harmonics and carrier harmonics, to understand the main loss factors. Next, the variation of the magnet eddy-current loss due to the magnet divisions is investigated from both results of the finite-element analysis and the theoretical solutions.

It is clarified that the magnet eddy-current loss is mainly produced by the carrier harmonics of the PWM inverter and that the axial length of the divided magnet should be smaller than the skin depth of the eddy currents produced by the major harmonics for the loss reduction.

II. CALCULATION METHOD

A. Outline of Calculation

In the cases of the vector-controlled motors, the known quantities for the finite-element analysis are the amplitude and phase of the armature current due to the controller command. The voltage waveform of the PWM inverter is unknown. The easiest way for the consideration of the carrier harmonics is to give the measured voltage or current waveforms to the analysis. However, this method is incompetent to estimate the characteristics before manufacturing the motor. Therefore, we developed a procedure whose outline is shown in Fig. 1 [4].

First, the nonlinear time-stepping finite-element analysis with the sinusoidal current due to the controller command is carried out as the precalculation. The main purpose of this analysis is to obtain the fundamental amplitude and phase of the armature voltage of the motor at the controlled condition. Thus, the rough time step is acceptable in this analysis. Accordingly, the theoretical voltage waveform of the PWM inverter, which corresponds to the calculated armature voltage, is generated by the wave generator simulating the pulse-modulation method of the inverter.

Next, the main calculation, which is the transient nonlinear time-stepping finite analysis coupled with the armature-voltage

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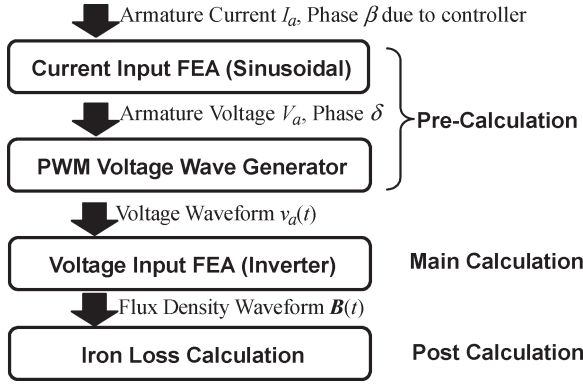


Fig. 1. Outline of calculation procedure.

equation, is carried out with the waveforms obtained by the precalculation. In this analysis, the time step should be short enough to estimate the kilohertz-order carrier harmonics.

Finally, the core and magnet losses are calculated from the time variation of the flux density and the eddy currents obtained by the main calculation. The Fourier transformation is applied to the flux-density and eddy-current waveforms at each finite element in order to classify the losses due to their origins.

B. Finite-Element Analyses

The formulation of the 3-D nonlinear time-stepping finite-element analysis is as follows:

$$\nabla \times \left(\frac{1}{\mu} \nabla \times \mathbf{A} \right) = \mathbf{J}_a - \sigma \left(\frac{\partial \mathbf{A}}{\partial t} + \nabla \phi \right) + \frac{1}{\mu_0} \nabla \times \mathbf{M} \quad (1)$$

$$\nabla \cdot \left\{ \sigma \left(\frac{\partial \mathbf{A}}{\partial t} + \nabla \phi \right) \right\} = 0 \quad (2)$$

where $\mathbf{A}$ and ϕ are the magnetic vector and the electric scalar potentials, μ and σ are the permeability and the conductivity, respectively. $\mathbf{J}_a$ is the armature-current density; $\mathbf{M}$ is the magnetization of the permanent magnet. The first-order tetrahedron edge-finite elements are applied in the discretization. The conductivity is only considered in the permanent-magnet region and neglected in the laminated-core region. In the precalculation in Fig. 1, $\mathbf{J}_a$ is given due to the controller command. On the other hand, in the main calculation considering the carrier, the following armature-voltage equation is coupled with (1) and (2):

$$v_a = v_{oa} + R_a i_a = \frac{d\Phi_a}{dt} + R_a i_a \quad (3)$$

where v_a and i_a are the armature voltage and current, respectively, R_a is the armature coil resistance, and Φ_a is the flux linkage of the armature coil. As shown in Fig. 1, v_a is given by the PWM voltage waveform generator. In this case, the dead time of the transistor is neglected.

C. Loss Calculation

Next, the core and magnet losses are calculated from the time variation of the flux density and the eddy currents obtained by (1) and (2).

TABLE I
SPECIFICATIONS OF ANALYZED MOTOR

Phases and poles	3phases, 8poles
DC voltage	400V
Carrier frequency	10kHz
Maximum rotational speed	10,000min ⁻¹
Maximum output	100kW
Diameter of stator and rotor	180mm, 123mm
Core length	130mm
Number of stator slot	48
Type of Magnet	Nd-Fe-B ($B_{remnant}=1.1T$)
Conductivity of Magnet	7.1×10^5 S/m
Steel plate for core	20RMHF1200, 0.2mm

The eddy-current loss of the magnet W_{mag} is calculated from the harmonic eddy-current densities as follows:

$$W_{mag} = \sum_n \left\{ \int_{\text{magnet}} \frac{|\mathbf{J}_n|^2}{2\sigma} dv \right\} \quad (4)$$

where $\mathbf{J}_n$ is the amplitude of the n th time-harmonic eddy currents.

On the other hand, the eddy-current and hysteresis losses W_e , W_h in the laminated core are calculated approximately from the harmonic flux densities [4] as

$$W_e = \sum_n \left\{ \int_{\text{iron}} K_e D(nf)^2 (B_{r,n}^2 + B_{\theta,n}^2) dv \right\} \quad (5)$$

$$W_h = \sum_n \left\{ \int_{\text{iron}} K_h D(nf) (B_{r,n}^2 + B_{\theta,n}^2) dv \right\} \quad (6)$$

where K_e and K_h are the experimental constants obtained by the Epstein frame test of the core material [4], D is the density of the electrical steel sheets, f is the fundamental frequency, $B_{r,n}$ and $B_{\theta,n}$ are the radial and peripheral components of the time-harmonic flux density at each finite element. Equation (6) is based on the approximation that the major and minor hysteresis loops are similar figures. This approximation will cause the calculation error. However, it can be considered as slight because the main component of the harmonic iron loss will be the eddy-current loss, which is in proportion to the square of the frequency (it will be discussed again in Section III-B).

In addition, when f_c/f (f_c is the carrier frequency of the inverter) is not an integer, the orders of the harmonics n are not integers either. In this case, the analysis due to (1)–(3) should be carried out during m time periods to make mf_c/f an integer.

III. RESULTS AND DISCUSSIONS

A. Analyzed Motor and Numerical Model

Table I shows the specifications of the analyzed motor. The motor has one interior magnet per pole. It is a sintered magnet whose remnant flux density is 1.1 T. One magnet is subdivided into 28 pieces along the axial length to prevent the eddy currents. The inverter is IGBT-type PWM inverter whose carrier frequency is 10 kHz.

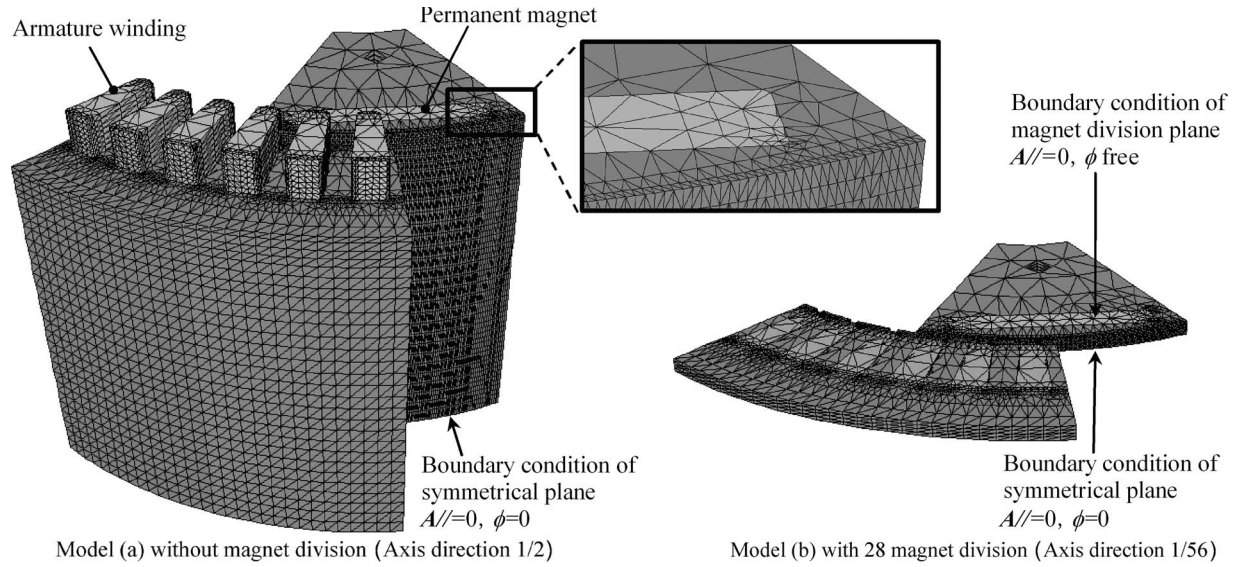


Fig. 2. Three-dimensional meshes.

TABLE II
DISCRETIZATION DATA

	Model (a) without magnet division	Model (b) with 28 magnet division (real case)
Number of time steps	1,024 steps / one time period of power supply	
Number of element (tetra)	427,350	49,368
Calculation time	1,291h / one time period	86h / one time period

Computer : Pentium4-3.6GHz PC

Fig. 2 shows the two kinds of 3-D finite-element meshes, which are models (a) and (b). Model (a) corresponds to the motor without the magnet division. In this case, the half region of the motor along the axis is discretized by the 3-D tetrahedron edge-finite elements with the boundary condition at the symmetrical side, i.e., setting the parallel component of $\mathbf{A}$ and ϕ as zero. Using this condition, the flux density is fixed as parallel and eddy current as normal to the plane. On the other hand, model (b) corresponds to the measured motor whose magnet is divided into 28 pieces. In this case, the analyzed region is reduced as 1/56 of core length, which corresponds to the half of the magnet, by imposing the different boundary conditions at each side. At the magnet-division side, only the magnetic vector potential $\mathbf{A}$ is set as zero. Using this condition, we can take the insulation between the divided magnets into account because both the flux density and the eddy current are fixed as parallel to the plane. In both models, the rotor region is shifted at each time step due to the rotational speed of the rotor.

Table II shows the discretization data of the main calculation considering the carrier harmonics. The number of the finite elements for model (b) is nearly 1/9 of model (a). Accordingly, the calculation time is 1/15. On the other hand, the calculation time of model (a) is over 1000 h.

B. Comparisons of Measured and Calculated Results

First, the calculated results of model (b) are compared with the measurement to verify the validity of the calculation method. In the measurement, the amplitude and phase of the armature current are set as 200 A, 50° constant.

Fig. 3 shows the measured- and theoretical-line-line-voltage waveforms and the included main harmonic components at 6000 min⁻¹. As shown in Fig. 1, the theoretical voltage waveform is generated by the wave generator simulating the pulse-modulation method of the inverter with the result of the precalculation. The orders of the major harmonics included in the PWM waveform are $(2i - 1)f_c/f$ plus-minus second, $(2i - 1)f_c/f$ plus-minus fourth, $2if_c/f$ plus-minus first, and $2if_c/f$ plus-minus fifth ($i = 1, 2, 3, \dots$), where f_c is the carrier frequency and f is the fundamental frequency. In the case of 6000 min⁻¹, f_c/f equals 25. Accordingly, the orders of the main harmonics are 25 plus-minus second, 25 plus-minus fourth, 50 plus-minus first, 50 plus-minus fifth, etc. It can be said that the theoretical waveform agree well with the measured one. The largest harmonics at this condition is 49th and 51st, whose frequency is 20 kHz. In detail, the harmonics included in the theoretical wave are slightly larger than the harmonics included in the measured wave because it neglects the dead time of the transistors.

Fig. 4 shows the measured and calculated iron losses including the magnet eddy-current loss. In the experiment, the iron loss is obtained by the subtraction of the output power and copper and mechanical losses from the input power [4]. The result of the precalculation, neglecting the carrier harmonics, is also shown for the comparison. The calculated result of the main calculation considering the carrier relatively agrees well with the measurement, whereas the result neglecting carrier underestimates the loss. The validity of the calculation is verified. It implies that the ratio of the magnet eddy-current loss to the total iron loss in case of this motor is small due to the effect of

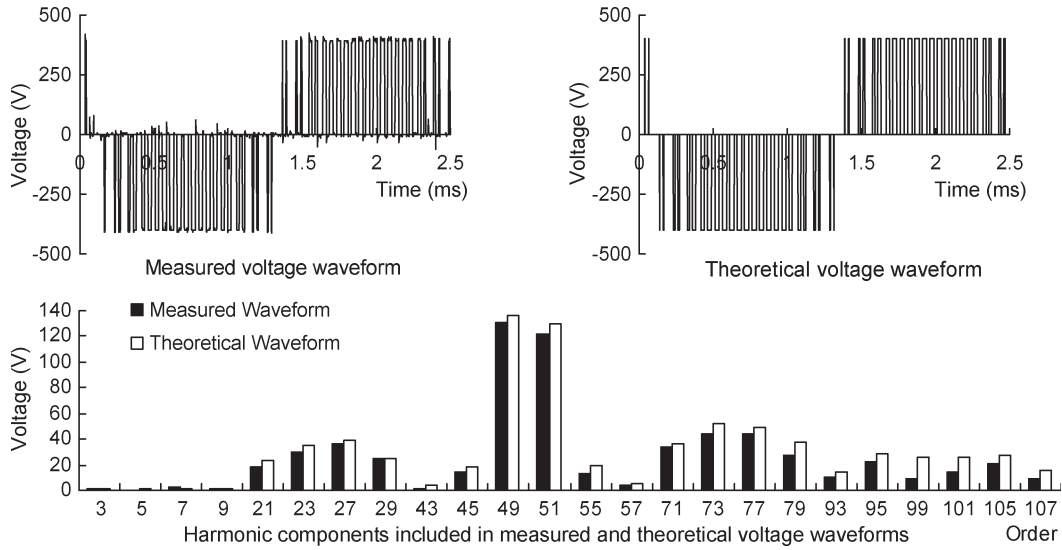


Fig. 3. Measured- and theoretical-line-line-voltage waveforms and included main harmonic components (6000 min^{-1}).

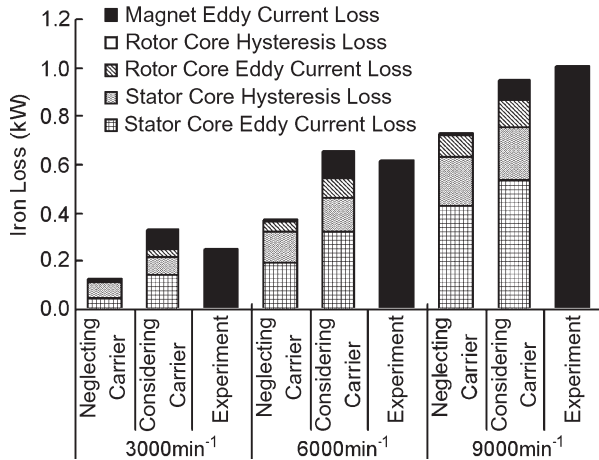


Fig. 4. Measured and calculated total iron losses.

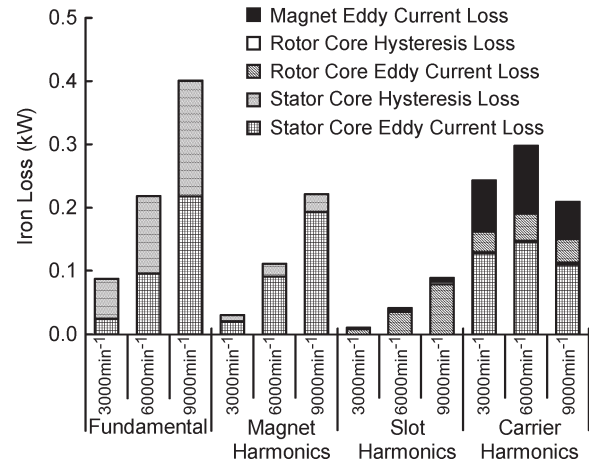


Fig. 5. Decomposed iron losses due to their origins.

the magnet divisions. However, it is still the main part of the loss generated at the rotor.

Fig. 5 shows the decomposed iron losses due to their origins, which are caused by the fundamental rotational field, harmonic magnetomotive forces of the rotor magnet, stator-slot harmonics, and carrier harmonics. They are classified from the harmonic eddy-current and hysteresis losses, which are calculated separately by (4)–(6) [4]. As we discussed in Section II-C, the results of the harmonic hysteresis losses calculated by (6) may include the error due to the simple approximation. However, the results of Fig. 5 imply that the harmonic hysteresis losses are very small as compared with the other harmonic losses. It can be said that the calculation error of the harmonic hysteresis loss is slight in the total iron loss.

On the other hand, Fig. 5 also shows that the magnet eddy-current loss is produced by both the stator-slot and the carrier harmonics and that it is mainly produced by the carrier harmonics. This must be the reason why the magnet eddy-current loss obtained by the calculation neglecting the carrier in Fig. 4 is very small. In addition, the magnet eddy-current loss produced by the carrier harmonics is largest at 6000 min^{-1} ,

whereas the other losses simply increase due to the rotational speed. It can be considered that the magnet eddy-current loss is mainly determined by the voltage-modulation ratio a of the PWM inverter. The voltage-modulation ratio a becomes nearly 0.6 at 6000 min^{-1} . At this condition, the major harmonics of $2f_c/f$ plus-minus first included in the voltage waveform becomes largest.

Figs. 6 and 7 show the magnet eddy-current distribution obtained by the precalculation neglecting the carrier and the main calculation considering the carrier, respectively. In the case of the result neglecting the carrier, it can be considered that the eddy currents are mainly produced by the stator-slot harmonics. Accordingly, their distribution varies due to the slot pitch, and they are concentrated at the sides of the magnet because the slot harmonic flux density is concentrated at the rotor surface [4]. On the other hand, the result considering the carrier shows remarkable difference. In this case, the eddy currents flow through the whole part of the magnet because they are mainly produced by the carrier harmonic flux density, whose distribution varies due to one pole pitch, and it enters into the rotor deeply [4]. Furthermore, the frequency of the major

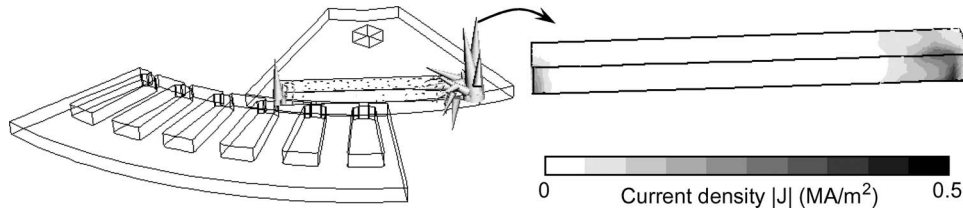


Fig. 6. Eddy-current distribution of magnet (28 magnet divisions, obtained by precalculation neglecting carrier, 6000 min⁻¹).

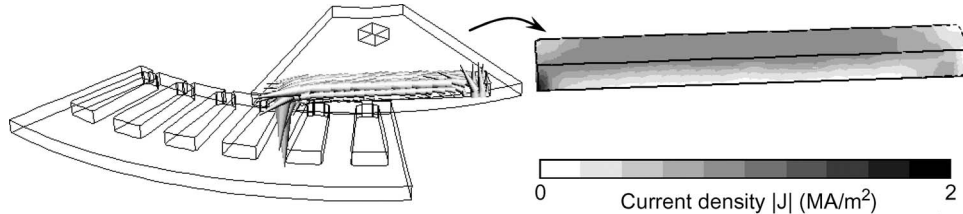


Fig. 7. Eddy-current distribution of magnet (28 magnet divisions, obtained by main calculation considering carrier, 6000 min⁻¹).

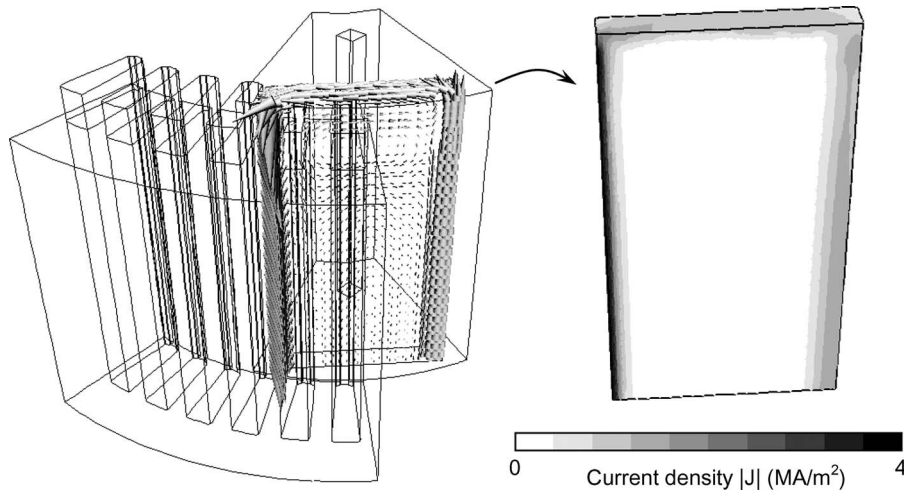


Fig. 8. Eddy-current distribution of magnet (without magnet division, obtained by main calculation considering carrier, 6000 min⁻¹).

carrier harmonics is higher than the slot harmonics. As a result, the eddy-current density of Fig. 7 is much higher than Fig. 6.

C. Variation of Magnet Eddy-Current Loss Due to Divisions

Next, let us investigate the effects of the magnet divisions. Fig. 8 shows the eddy-current distribution of model (a), which is without the magnet division, obtained by the main calculation considering the carrier. In this case, the eddy currents concentrate at the edge of the magnet due to the skin effect. The maximum eddy-current density is nearly twice of the case of 28 magnet divisions. On the other hand, it can be seen that the region where the eddy currents flow is restricted.

Fig. 9 shows the variation of the calculated total iron loss due to the number of the magnet divisions along the axial length. The variation of the losses generated at the stator and rotor cores is slight. On the other hand, the magnet eddy-current loss becomes maximum at eight divisions. In this case, the magnet eddy-current loss becomes nearly three times as large as the loss of the measured motor, whose magnets are divided into 28 pieces.

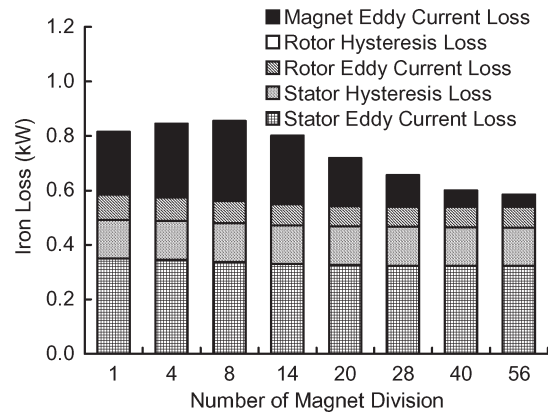


Fig. 9. Variation of losses due to axial magnet divisions.

Fig. 10 shows the classified magnet eddy-current losses due to their origins. It indicates that the loss produced by the carrier becomes maximum at eight divisions, whereas the loss produced by the stator-slot harmonics simply decrease due to the increase of the divisions.

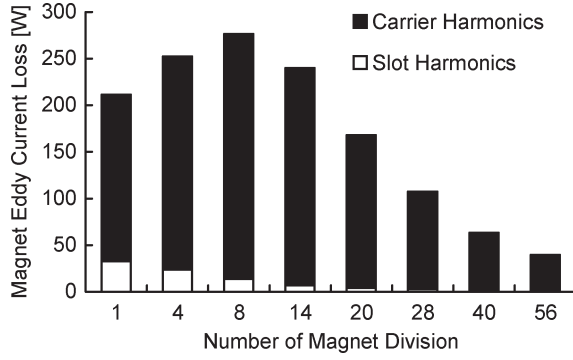


Fig. 10. Variation of classified eddy-current losses due to magnet divisions.

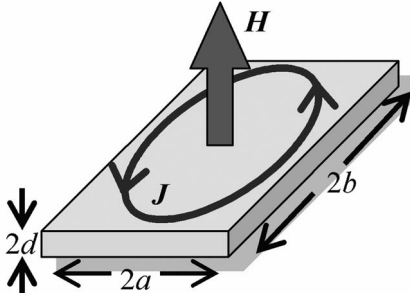


Fig. 11. Simple model for theoretical solution of eddy-current loss.

To understand these results, let us introduce the theoretical solutions of the eddy-current loss in thin conductor when the uniform magnetic field is applied [5], i.e.,

$$\begin{aligned}
 P &= \frac{2d}{\sigma} \int_0^b \int_0^a |\mathbf{J}|^2 dx dy \\
 &= \frac{2bd}{\sigma \delta} |H|^2 \frac{\sinh \frac{2a}{\delta} - \sin \frac{2a}{\delta}}{\cosh \frac{2a}{\delta} + \cos \frac{2a}{\delta}} - \frac{128a^4 d}{\pi^5} \frac{|\gamma^2 H|^2}{\sigma} \\
 &\quad \times \sum_{n=0}^{\infty} \frac{(\lambda_n^2 - 2\beta_{nr}^2) \beta_{nr} \lambda_n^3 \sinh 2\beta_{nr} b}{(2n+1)^5 |\beta_n|^6 (\cosh 2\beta_{nr} b + \cos 2\beta_{ni} b)} \\
 &\quad - \frac{128a^4 d}{\pi^5} \frac{|\gamma^2 H|^2}{\sigma} \\
 &\quad \times \sum_{n=0}^{\infty} \frac{(\lambda_n^2 + 2\beta_{nr}^2) \beta_{ni} \lambda_n^3 \sin 2\beta_{ni} b}{(2n+1)^5 |\beta_n|^6 (\cosh 2\beta_{nr} b + \cos 2\beta_{ni} b)} \quad (7)
 \end{aligned}$$

where a , b , and d are the half of the width, length, and thickness of the conductor as shown in Fig. 11, δ is the skin depth, and

$$\lambda_n = (2n+1) \frac{\pi}{2a} \quad (8)$$

$$\gamma = \frac{1+j}{\delta} \quad (9)$$

$$\beta_n = \beta_{nr} + j\beta_{ni} = \sqrt{\lambda_n^2 + \gamma^2} \quad (10)$$

When the skin depth δ is large enough as compared with a and b , (7) becomes

$$P \cong \frac{2a^3 b d \sigma}{3} |\mu H|^2 + \frac{128a^4 d \sigma}{\pi^5} |\mu H|^2 \sum_{n=0}^{\infty} \frac{\tanh \lambda_n b}{(2n+1)^5} \quad (11)$$

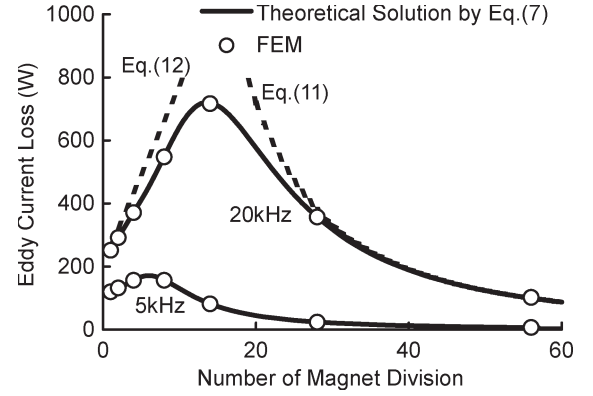


Fig. 12. Theoretical solution of thin-plate eddy-current loss.

On the other hand, when δ is much smaller than a and b , (7) is approximated as

$$P \cong \frac{2bd}{\sigma \delta} |H|^2 + \frac{2ad}{\pi^2 \sigma \delta} |H|^2 \sum_{n=0}^{\infty} \frac{8}{(2n+1)^2} = \frac{2bd}{\sigma \delta} |H|^2 \frac{b+a}{b} \quad (12)$$

In this expression, the meaning of $2bdH^2/\sigma\delta$ is the eddy-current loss of the infinitely long conductor between $2b$ length. It corresponds to the result of the 2-D analysis. Thus, the meaning of (12) can be considered as the compensated value due to width a and length b . It implies that the eddy-current loss is simply determined by the total edge length of the conductor due to the concentration of the eddy current when the skin effect is strong.

Fig. 12 shows the eddy-current loss due to the number of the conductor divisions calculated by (7), (11), and (12) when the imposed uniform flux density μH is 10 mT, and the frequencies are 5 and 20 kHz, which nearly correspond to the major slot and carrier harmonics in the motor, respectively. The result obtained by the finite-element analysis under the same condition is also shown for the comparison. The result of the finite-element analysis is almost identical to the solution of (7). The results also indicate that the loss becomes maximum when the width of the conductor $2b$ is nearly twice of the skin depth δ . As a result, the maximum point is shifted due to the frequency of the magnetic field. When the number of the magnet divisions is smaller than this point, the eddy-current loss can be approximated by (12). Accordingly, the eddy-current loss increases due to the divisions because the total edge length increases. On the other hand, when the number of the magnet divisions is large, the eddy-current loss can be approximated by (11), which results in the decrease of the eddy-current loss due to the divisions.

Fig. 13 shows the variation of the eddy-current path at 20 kHz, which corresponds to the major carrier harmonics in the motor. In this case, the skin depth is 4.1 mm. The maximum vector of the eddy currents at 8 and 14 divisions are nearly the same as no division, whereas the total edge length increases. On the other hand, in the case of 28 divisions, the eddy currents decrease because the length of the magnet is smaller than the twice of the skin depth.

From these results, it can be said that the axial length of the divided magnet of the interior permanent-magnet motor should

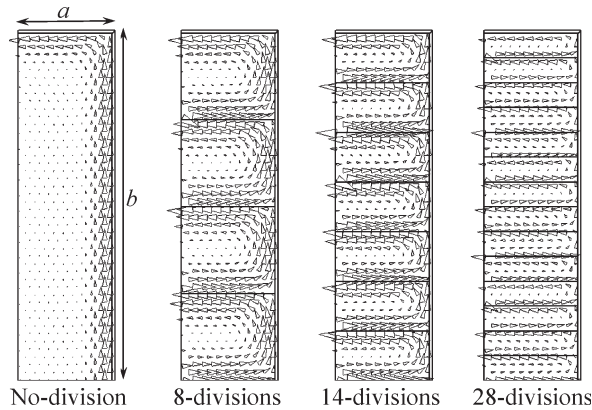


Fig. 13. Increase of eddy-current path due to divisions.

be smaller than twice of the skin depth of the eddy currents produced by the major harmonic field for the effective loss reduction.

IV. CONCLUSION

The losses of the interior permanent-magnet motors considering the magnet eddy currents and carrier harmonics of the PWM inverter are investigated by the 3-D nonlinear time-stepping analysis and the theoretical solution of the eddy-current loss. The measured and the calculated results of the total iron loss including carrier harmonics agree well. The magnet eddy-current loss of the analyzed motor is mainly produced by the carrier harmonics. From the results of the finite-element analysis and the theoretical solution of the eddy-current loss, it is concluded that the axial length of the divided magnet of the interior permanent-magnet motor should be smaller than twice of the skin depth of the eddy currents produced by the major harmonic field for the effective loss reduction.

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